Vertical shifts and dilations of sine and cosine functions (Extended)

Core Concepts: Vertical shifts and dilations of sine and cosine functions (Extended)

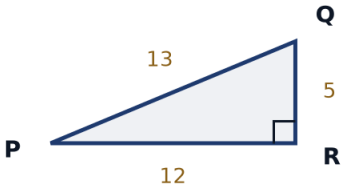
1 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(2)$.

a. 19

b. 25

c. 12

d. 23



2 Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

3 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(3)$.

a. 25

b. 31

c. 18

d. 29

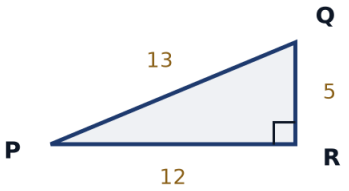
5 Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$



6 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

8 Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

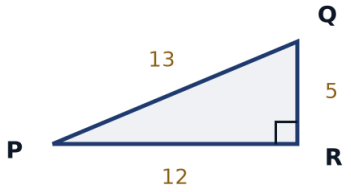
9 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5



Exam-Style Problem Solving & Modeling

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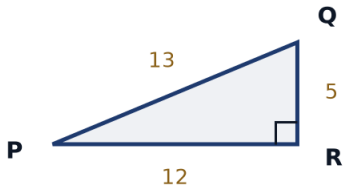
10 For the model $f(x) = 6x + 7$ in vertical shifts and dilations of sine and cosine functions (extended), evaluate $f(5)$.

a. 37

b. 43

c. 30

d. 41



11 Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

12 A data set recorded for vertical shifts and dilations of sine and cosine functions (extended) consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

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